



Cosmology

The physics of the universe

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Overview

What is Cosmology?

Our Place in the Cosmos

The Big Bang

Dark Matter

Dark Energy

Inflation & The CMB

Modern Astronomy

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- Age: How old is the universe?
- Birth: Is there a beginning of time? What was it like?
- Fate: Will everything come to an end?
- Size: How big is the cosmos? Is there an edge?
- Shape: What happens as you travel across the universe?

Our Place in the Cosmos

Earth



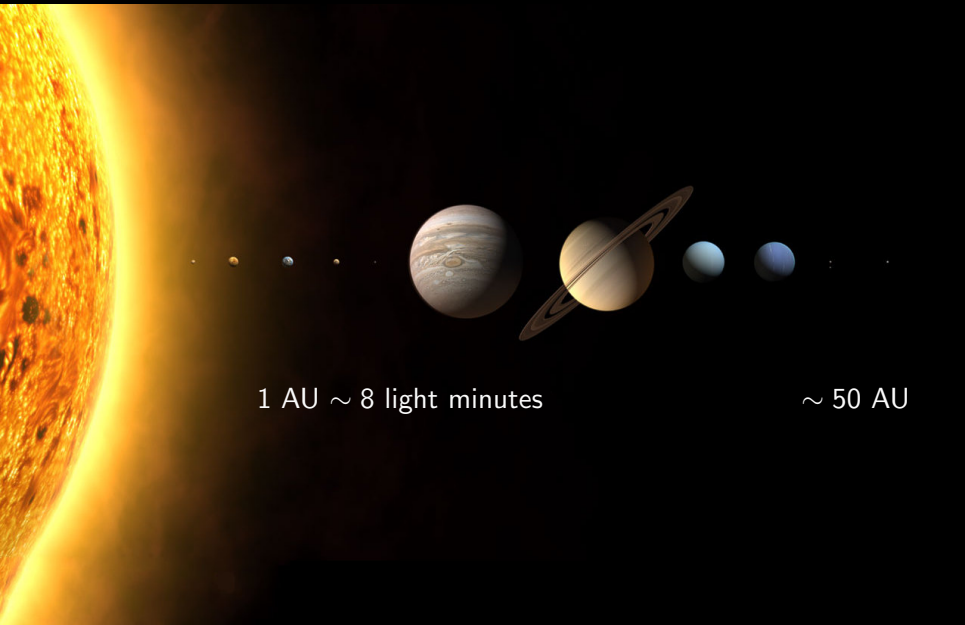
Our Place in the Cosmos

Stars



Our Place in the Cosmos

The Solar System



Our Place in the Cosmos

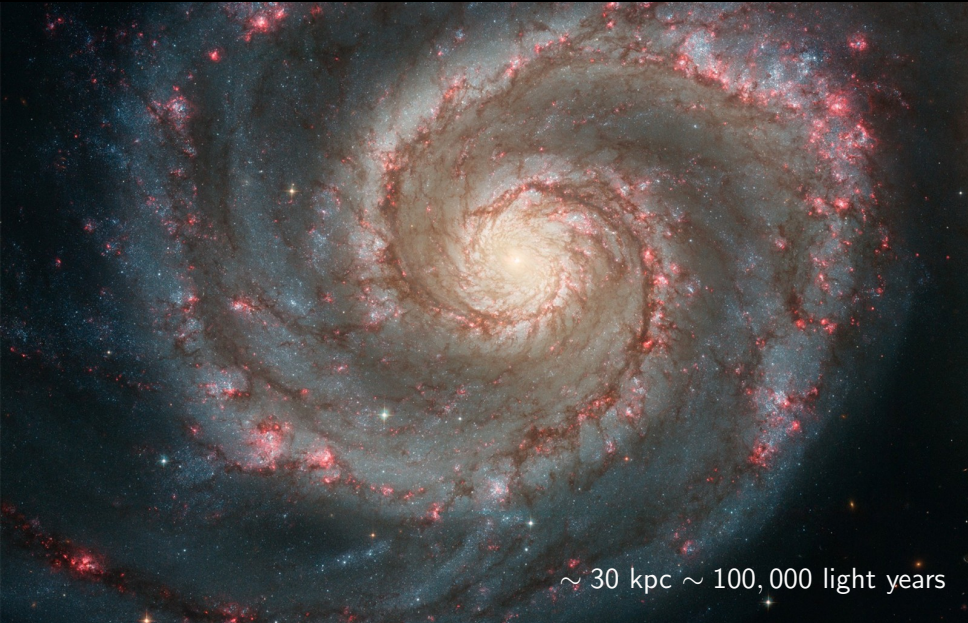
Star Clusters



$\sim \text{pc} = \text{parsec} \sim 3 \text{ light years}$

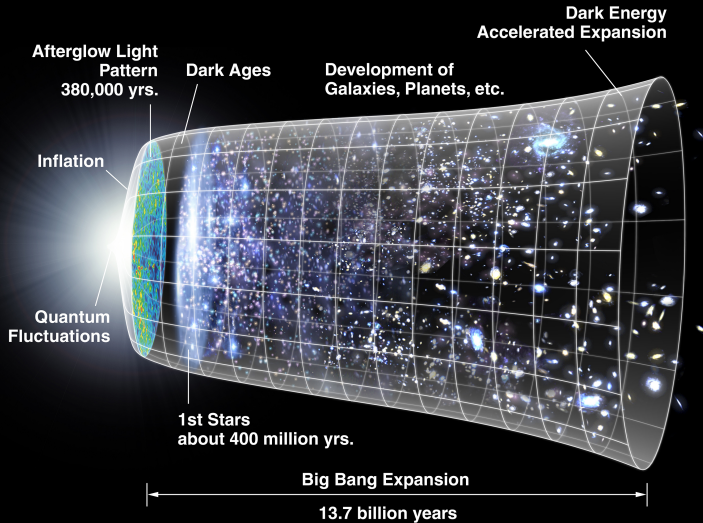
Our Place in the Cosmos

Galaxies



$\sim 30 \text{ kpc} \sim 100,000 \text{ light years}$

The Big Bang



The Big Bang

Evidence

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- ▶ The Cosmic Microwave Background Radiation

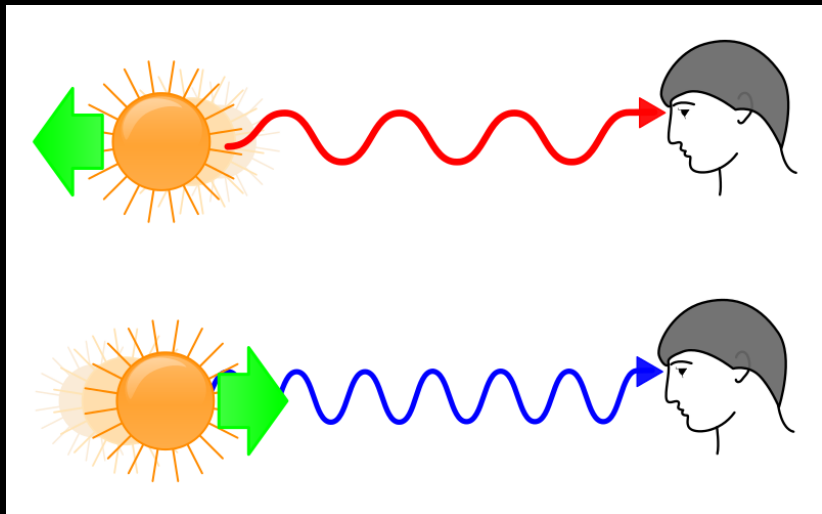
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2. The Further away they are, the faster they are moving away

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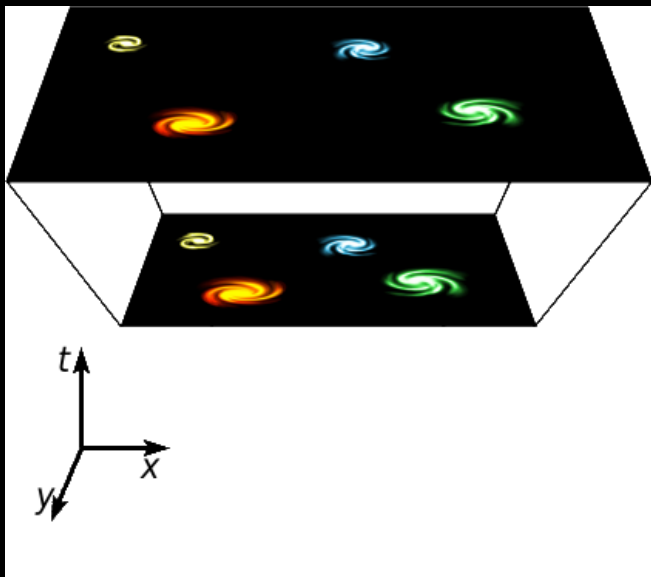
Galactic Redshift

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1. All Galaxies are moving away from us
2. The Further away they are, the faster they are moving away
3. If we were to go to another galaxy, we would see exactly the same thing

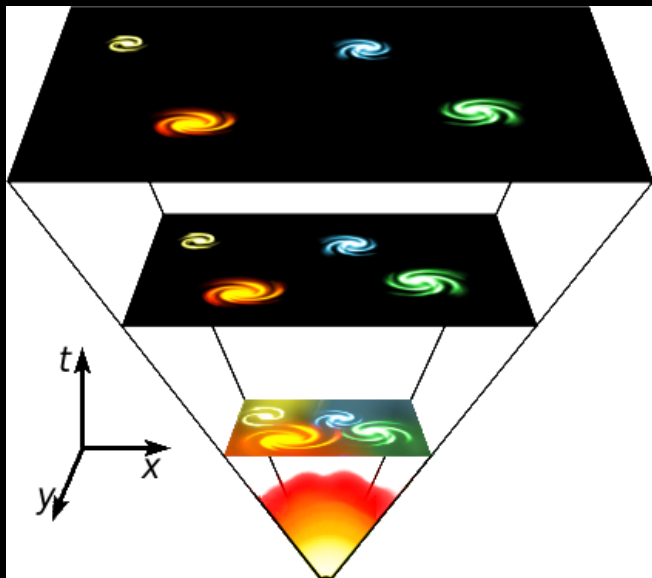
The Big Bang

A stretching universe:



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- ▶ This is best explained as an 'afterglow' of the Big Bang.

The Big Bang

Common Misconceptions

The Big Bang does *NOT* look like this:



The Big Bang

Common Misconceptions

- ▶ Despite the name, the Big Bang is not an explosion.

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- ▶ Despite the name, the Big Bang is not an explosion.
- ▶ The Universe is not necessarily expanding into anything.

Current Issues in Cosmology

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- ▶ $t = 0$

Dark Matter

The Problem



Dark Matter

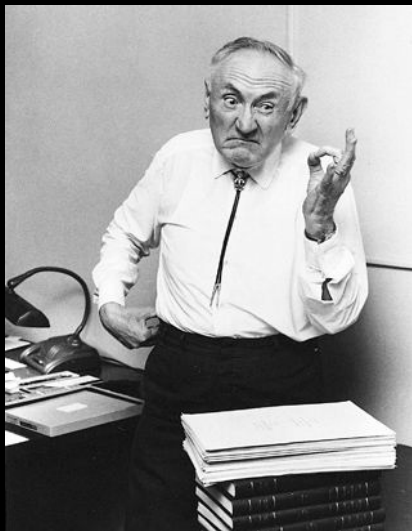
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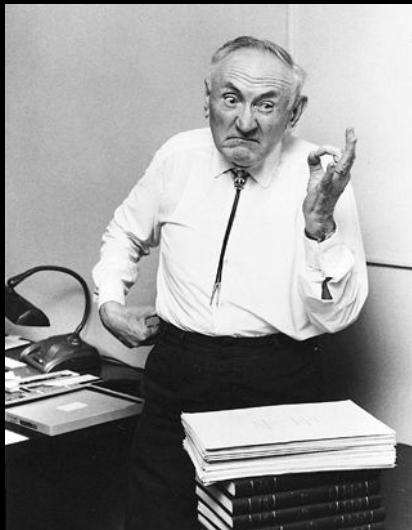
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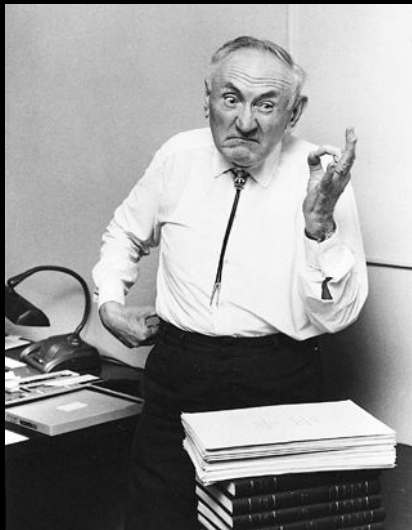
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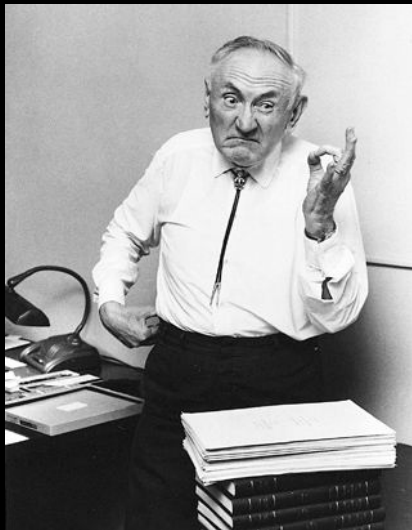
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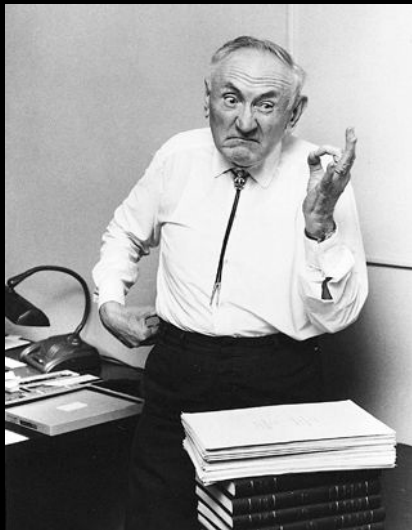
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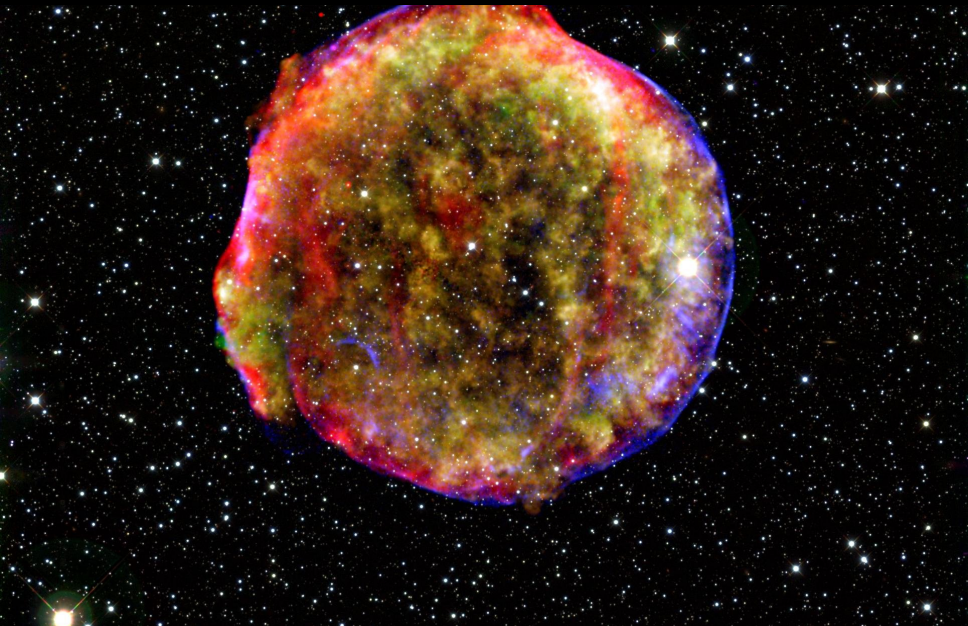
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- ▶ Right now it's really just a name for a phenomenon we don't understand yet.



Dark Energy

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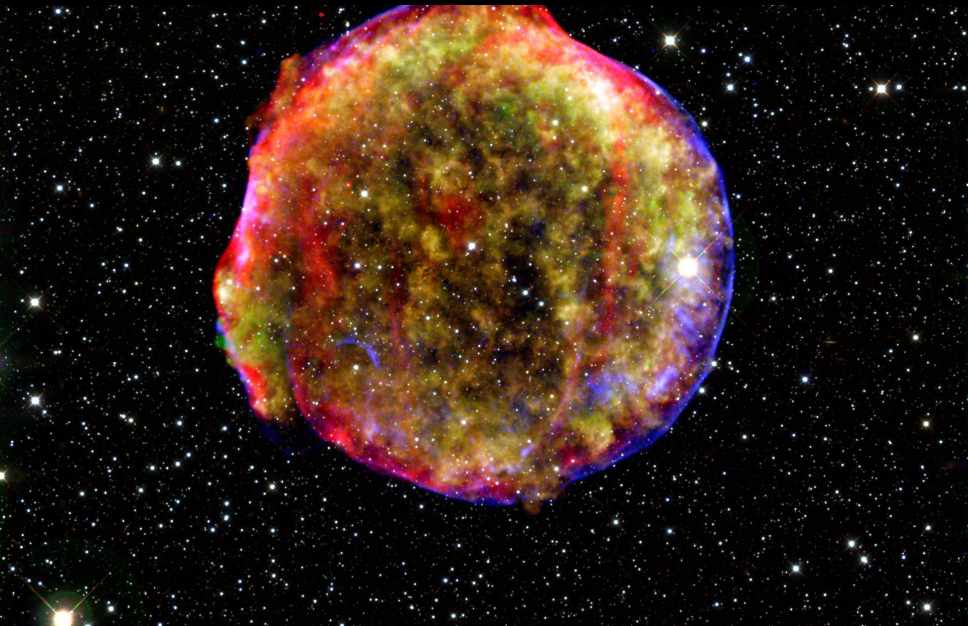
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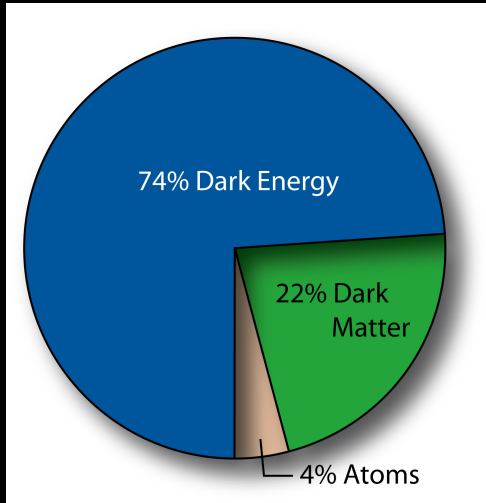
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- ▶ One of the ways in which this could happen would be the presence of an energy permeating the universe, accelerating the expansion.
- ▶ We call this Dark Energy
- ▶ Like Dark Matter, it's really just a name for a phenomenon which we observe, and haven't been able to explain yet

The Composition of the Universe



The Cosmic Microwave Background Radiation (CMB)

Our window to the early universe

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Our window to the early universe

- ▶ The CMB is one of the primary observational tools in modern Cosmology.

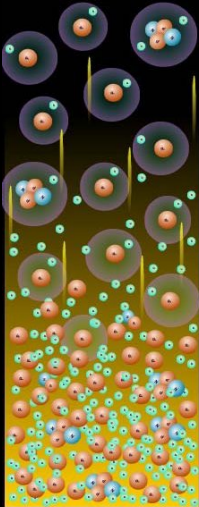
The Cosmic Microwave Background Radiation (CMB)

Our window to the early universe

- ▶ The CMB is one of the primary observational tools in modern Cosmology.
- ▶ It enables us to look at the universe when it was a mere 380,000 years old.

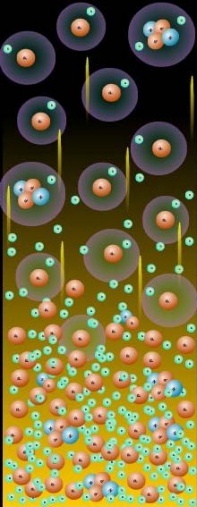
The surface of last scattering

“Let there be light”



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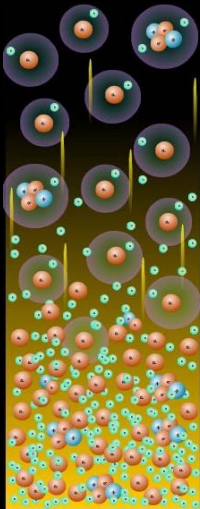
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- At early times the universe was very dense.

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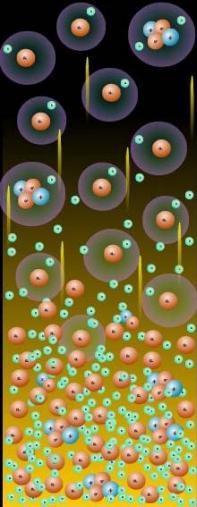
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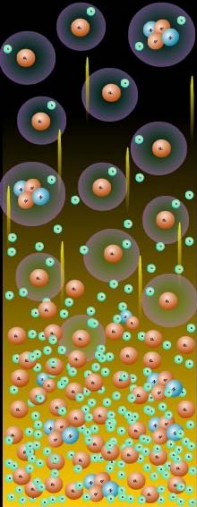
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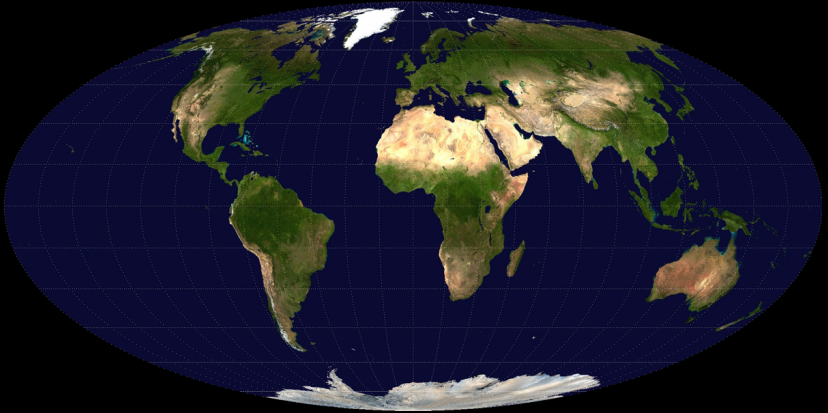
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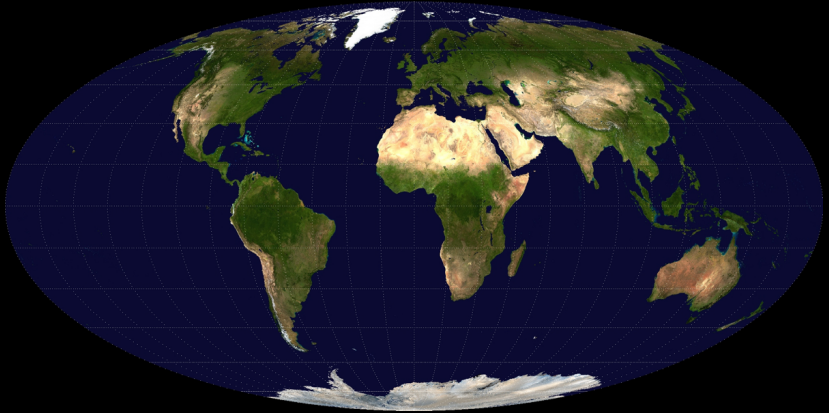
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- ▶ Light could not travel through this opaque plasma.
- ▶ At some point ≈ 3000 yrs after the big bang, the universe transitioned to being transparent to light.
- ▶ This light was now able to travel (essentially) unimpeded, and this “afterglow” is what we now call the CMB.

Anisotropies in the CMB



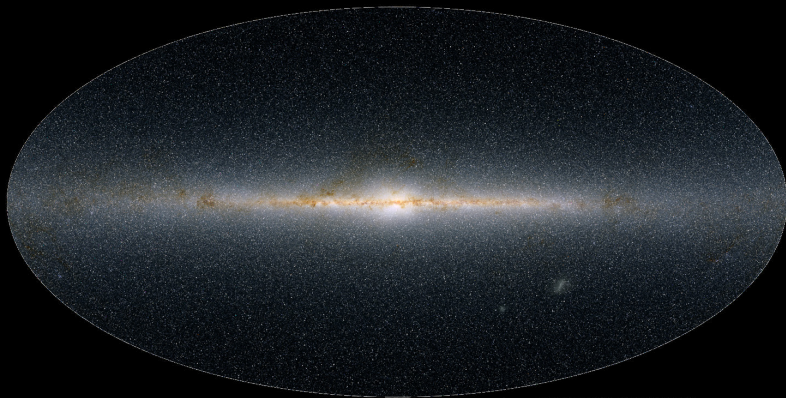
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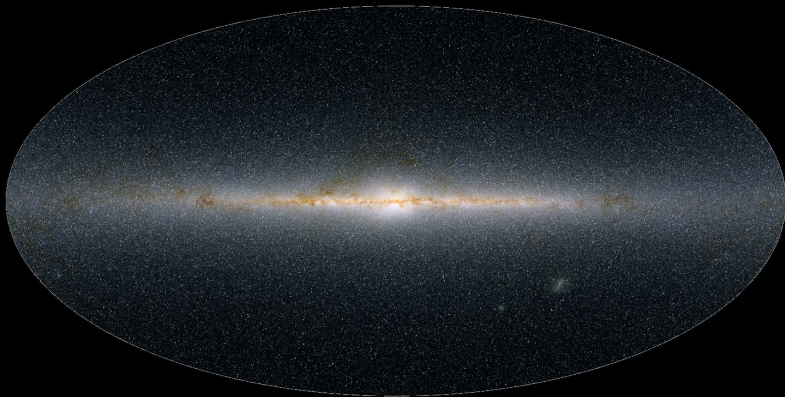
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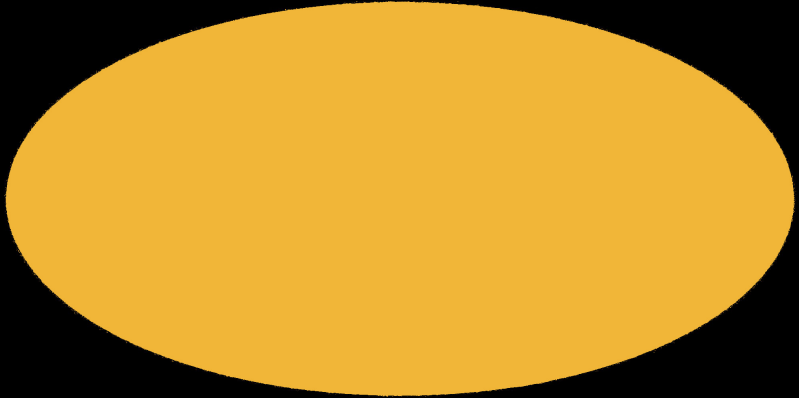
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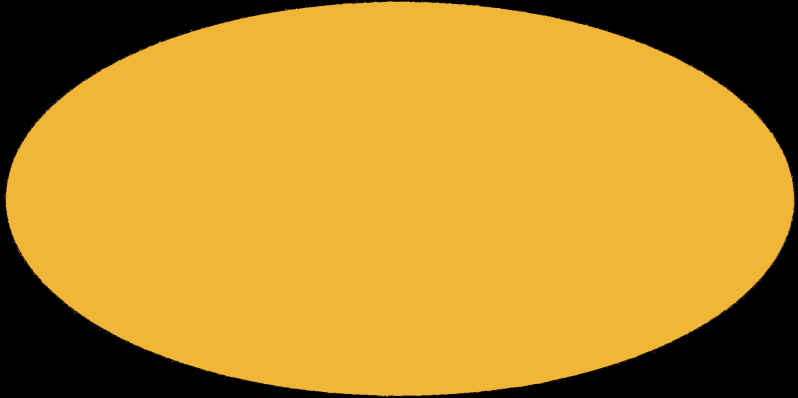
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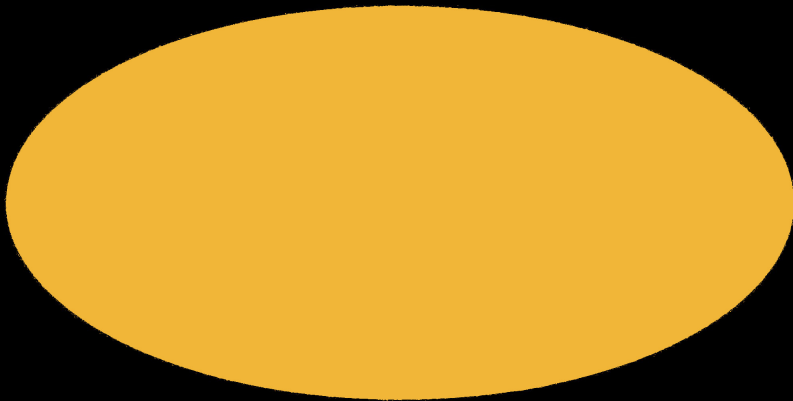
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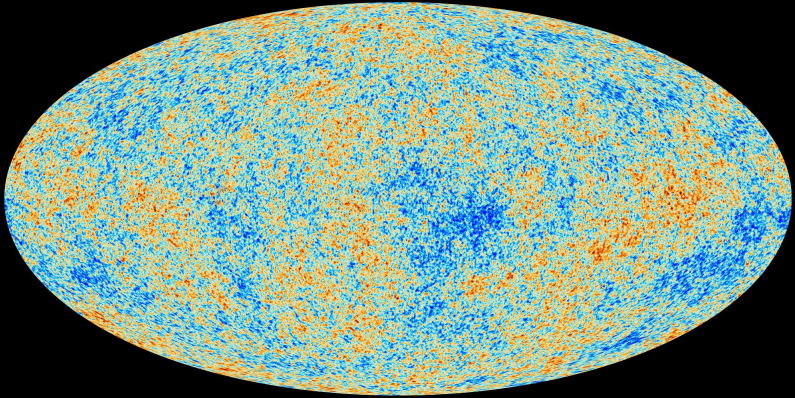
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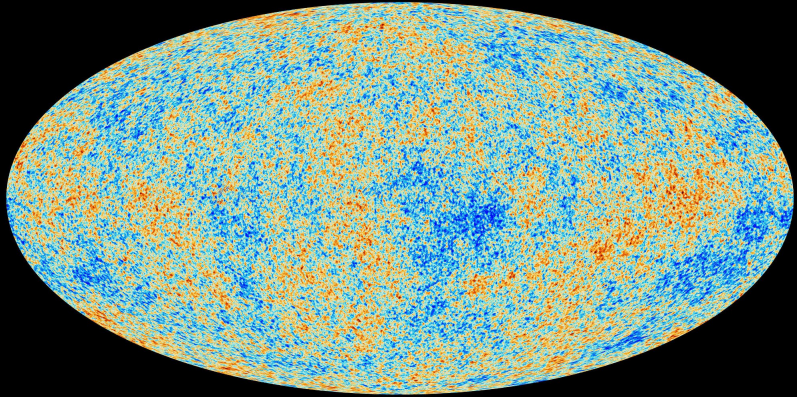
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Anisotropies in the CMB



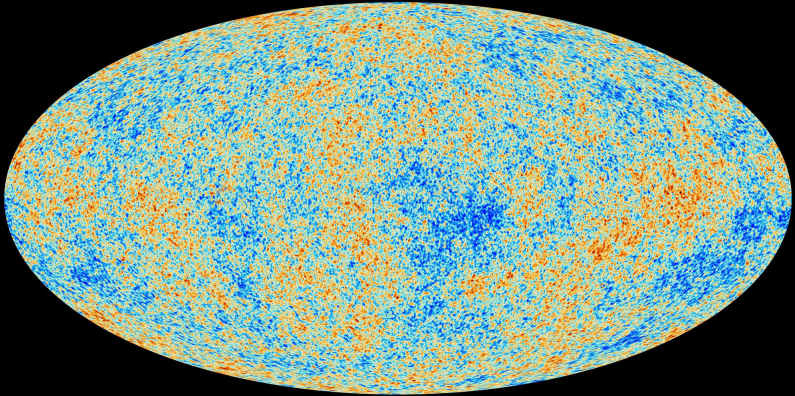
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Anisotropies in the CMB



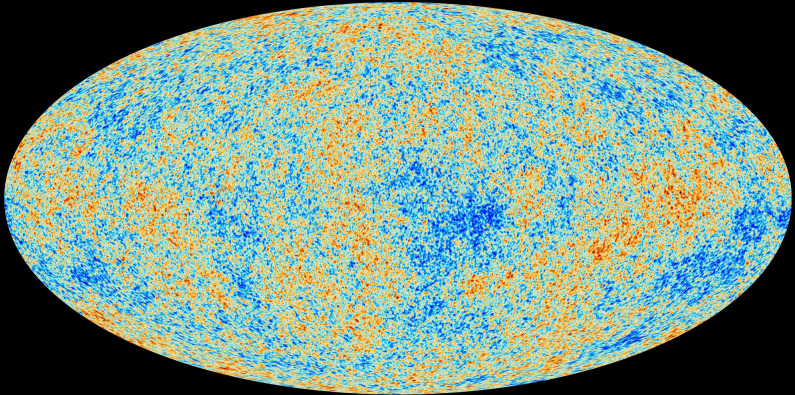
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- ▶ The *Theory of Inflation* predicts these inhomogeneities,
- ▶ Which are created by *quantum fluctuations*.
- ▶ These are the seeds of structure formation in the universe.

The dawn of a new era in astronomy

The rise of the mega-telescope

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- ▶ Square Kilometre Array

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- ▶ Planck Satellite

Square Kilometre Array (SKA)

2019: World-wide radio telescope



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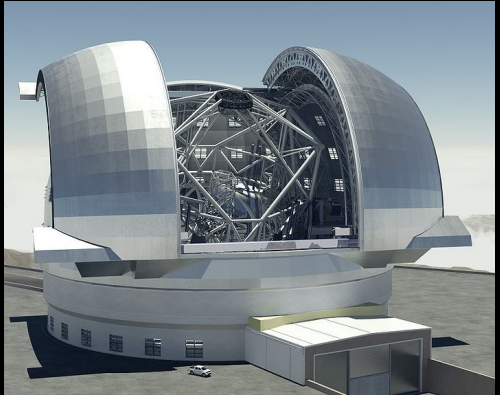
2019: World-wide radio telescope



- ▶ Cost: €1.5bn
- ▶ Testing GR, DE, DM, Magnetism...
- ▶ Imaging Hydrogen through the universe will build a 3D picture of the first ripples of structure.

European Extremely Large Telescope (E-ELT)

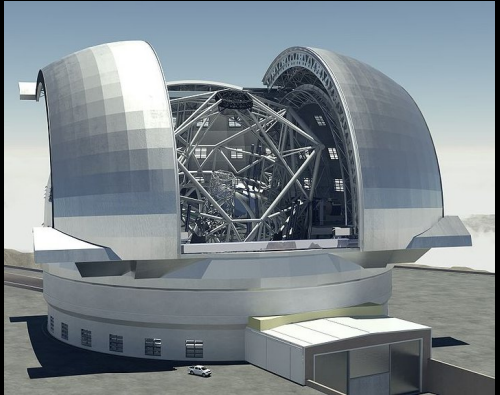
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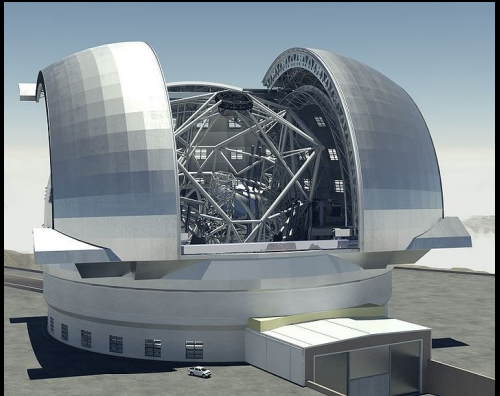
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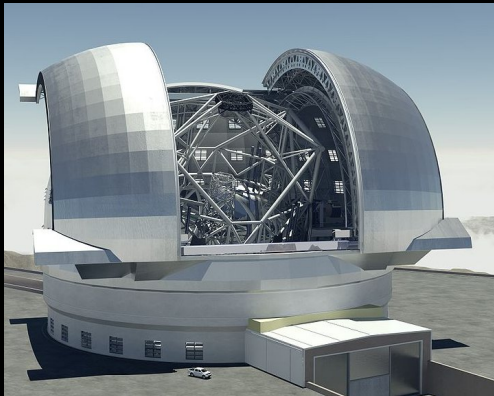
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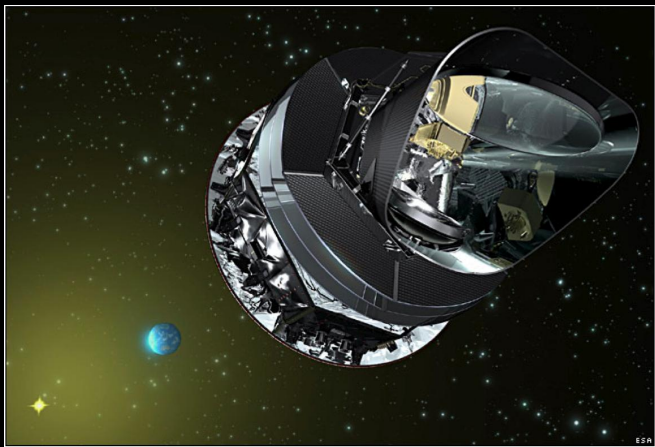
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- ▶ Cost $\approx$ €1.3bn
- ▶ Search for extrasolar planets.
- ▶ Over a 20 year baseline, it will be sufficiently accurate to watch the universe expand.



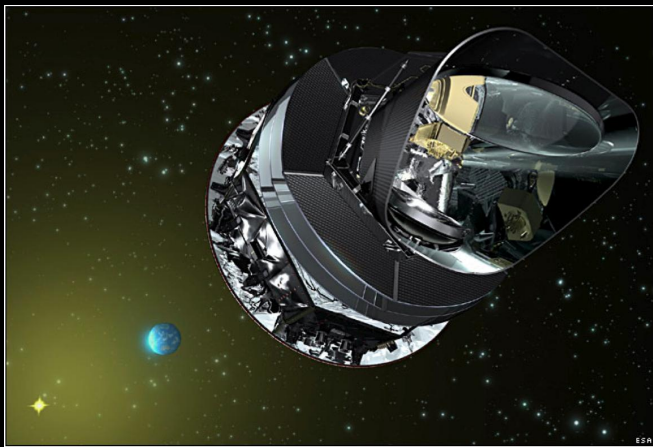
The Planck Satellite

In operation: Space-based microwave telescope



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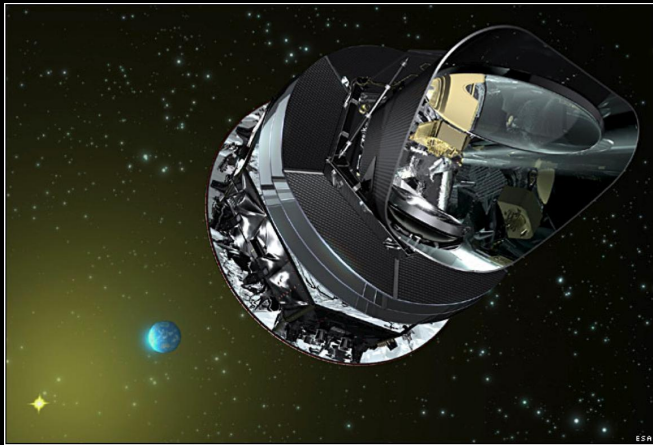
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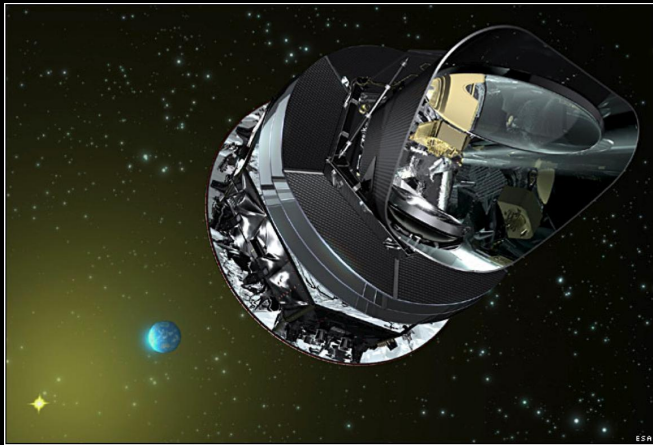
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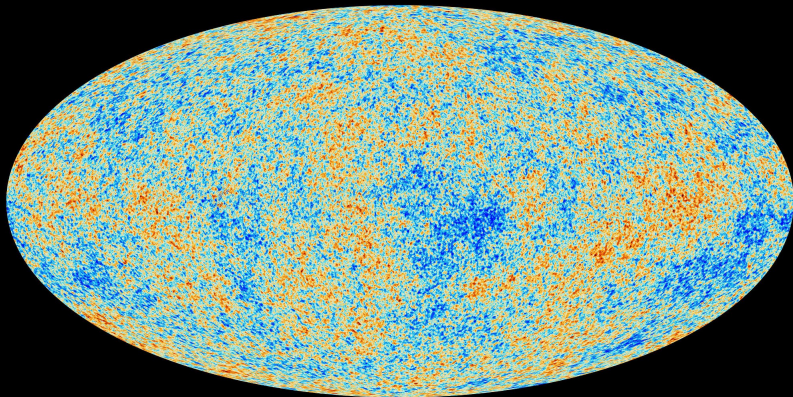
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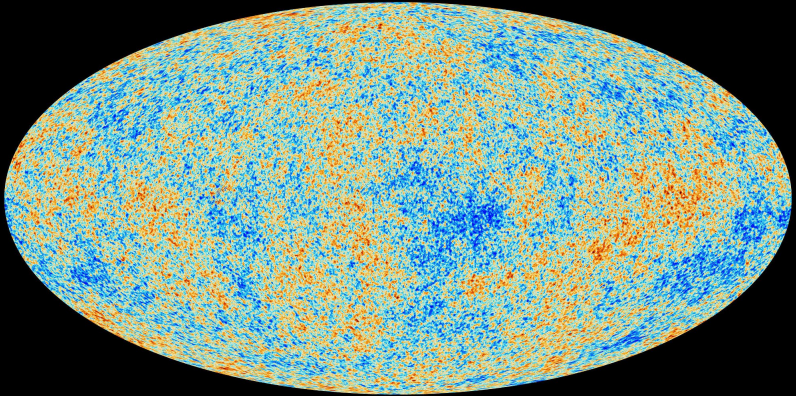


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- ▶ Cataloguing galaxy clusters
- ▶ Providing essential data for accurate exploration of cosmological theories

Summary

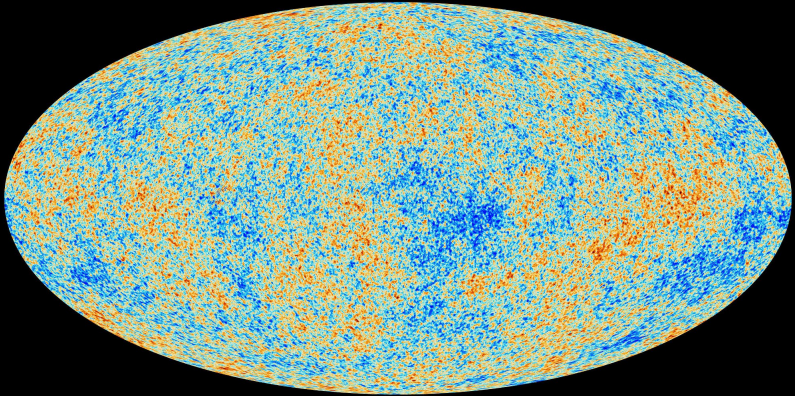


Summary



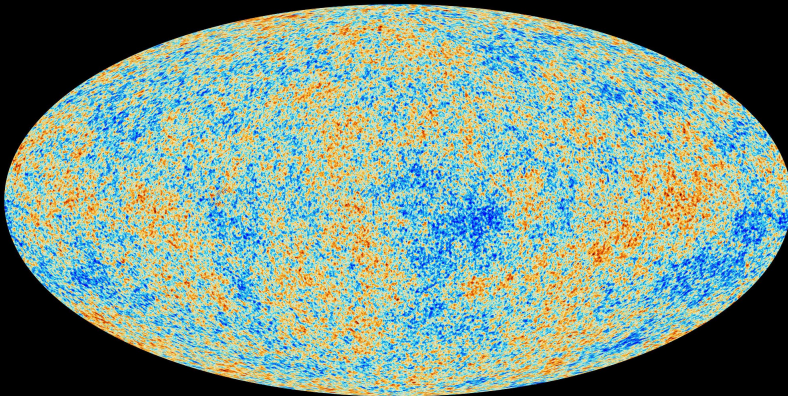
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- ▶ There are many unsolved theoretical problems,
- ▶ And a wealth of data to explore them with.